
Impact of the Transition from Caged to Cage-free Housing on the Welfare of Laying Hens

Cynthia Schuck-Paim & Wladimir J. Alonso

Introduction

In Chapter 1 [1] we present an analytical framework to quantify the loss of welfare experienced by animals in an explicit and comparative manner, based on estimates of the intensity and duration of the ‘pain’ (used as an operational term for affective states of negative valence) endured as a result of different welfare challenges (of both physical and psychological nature) experienced over their lives. By mapping aversive experiences onto a common scale with real-world meaning (time in pain), the framework enables the objective comparison of the negative affective experiences endured by animals as a result of practices and conditions of very different nature.

Next, we applied the framework to estimate the time laying hens spend in pain as a result of each of a number of injuries (Chapters 3 [2], 4 [3], 7 [4]), diseases (Chapter 5 [5]) and deprivations of motivated behaviors (Chapter 6 [6]) over the laying cycle. Our goal was to provide a quantitative basis upon which to evaluate the impact of reforms aimed at improving the welfare of commercial layers, specifically the move away from conventional caged systems to cage-free housing currently underway in many countries. To this end, we analysed the welfare challenges encountered over the laying period in conventional cages, furnished cages and cage-free aviaries (single- and multi-tier aviaries). Instead of considering an exhaustive list of every welfare challenge affecting commercial layers during this phase, we focused on those offenses most likely to be affected by the transition from conventional cages

(If you prefer to have this material in e-book format or printed format, please find it at <https://www.amazon.com/Quantifying-Pain-Laying-Hens-comparative-ebook/dp/B09BR69NV6/>)

Please cite it as

Schuck-Paim C., Alonso WJ 2021. Impact of the Transition from Caged to Cage-free Housing on the Welfare of Laying Hens. In: Schuck-Paim C, Alonso WJ, editors. Quantifying Pain in Laying Hens. Independently published. <https://tinyurl.com/bookhens>

A blueprint for the comparative analysis of pain

to cage-free aviaries (see Chapter 2 [7] for a detailed justification of the scope of analysis and criteria for the choice of welfare challenges).

In this chapter, we focus on the quantification of the burden of pain experienced by laying hens kept in these different housing systems when the cumulative load of all welfare challenges examined is considered. To this end, we estimate the total time spent in four categories of pain intensity as our basis for comparison (for a definition of the categories, see Box 1.2 in Chapter 1 [1]), enabling the assessment of the impact of the reform along a scale of negative experiences, all grounded on explicit parameters with biological meaning. For example, while some policies and reforms may have a large impact on less severe pain states, others can mitigate the most extreme (excruciating) pain experiences. By being able to differentiate these effects, the pursuit of different priorities becomes possible.

In the next section we discuss the relative contribution of each one of the welfare challenges analysed to the total burden of pain endured by the average member of a flock, to only then explore the compound effect of all challenges in the different housing systems. For a thorough justification of the time in pain that individuals endure as a consequence of each welfare offense, we refer the reader to the specific chapters addressing each condition, where the evidence supporting the parameter values and estimates is discussed. Because our interest is in the experience of layer populations in commercial settings, estimates of the prevalence of each condition were based on data from commercial flocks only (Chapter 8 [8]).

As demonstrated in a recent meta-analysis [9], management experience and the degree of technological maturity of a production system can act as important confounders of the prevalence of different welfare problems in comparisons of production conditions (e.g. some welfare challenges may see their prevalence temporarily increased following a transition to a new system, before knowledge and experience accrue [9,10]). However, as discussed in Chapter 8 [8], the data needed to control for differences in these factors is not yet available in most cases, so the estimates are most likely to reflect a situation where the technological, genetic and managerial maturity of cage-free housing is not yet on par with that already achieved by caged systems. Accordingly, the estimated time in pain averted with the transition to cage-free housing is likely to be higher than that presented in the next sections once the transition is more consolidated.

All parameter values used for the analysis (estimates of duration, intensity, number of occurrences and the prevalence of each welfare offense) are

justified in previous chapters. However, we invite the reader to examine the extent to which the estimates presented here are sensitive to our choices and assumptions, at the platform <https://pain-track.org/hens>. In this web application, all parameter values can be altered, with the immediate update of all estimates.

Relative burden of pain in different housing systems

Figure 9.1 shows the ‘relative’ contribution of each welfare challenge to the total time spent in each category of pain intensity by the average flock member (an estimate that takes into account the prevalence of the challenge in the population). The discussion that follows in this section is based on the estimates shown in this figure.

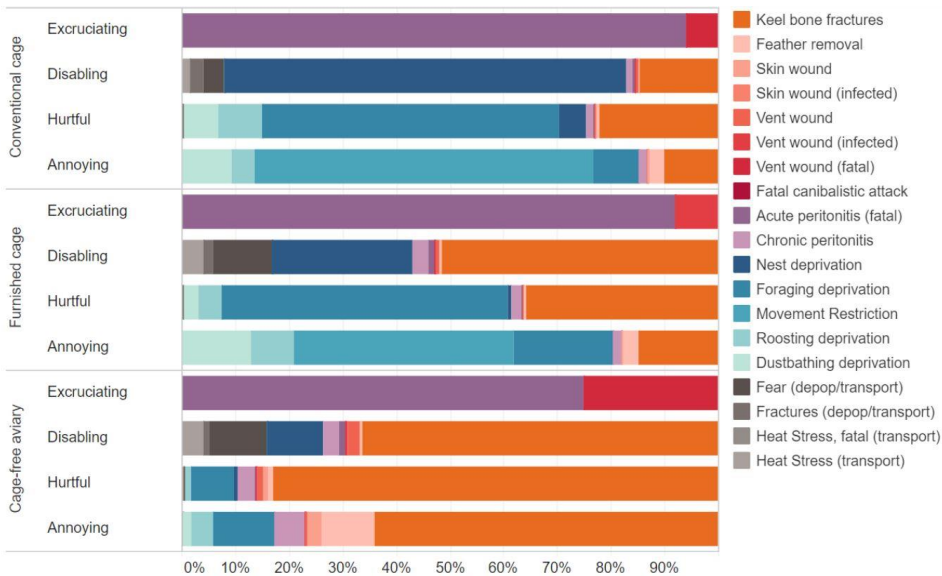


Figure 9.1. Contribution of each harm **relative** to the total estimated time (scaled to 100%) endured **by the average flock member (i.e. weighted by prevalence)** in the four categories of pain intensity. Here, the absolute size of each colored bar should not be compared across intensities or systems (see <https://welfarefootprint.org/research-projects/laying-hens> for a dynamic figure that enables inspecting values).

Estimates of the absolute time spent in pain differed across intensity categories and housing conditions, but in Figure 9.1 total times in pain are scaled to 100% to enable a clearer appreciation of the main drivers of pain in each case before exploring absolute differences among the three housing

A blueprint for the comparative analysis of pain

conditions. To compare absolute times in pain across intensity categories or housing systems, please refer to [figures 9.2](#) and [9.3](#).

The estimates in [Figure 9.1](#) confirm the expectation that keel bone fractures (orange bars) are among the most serious welfare problems affecting laying hens in modern egg production, and the main cause of physical pain. Keel fractures account for most of the time spent in disabling pain in cage-free aviaries (66%) and in furnished cages (51%) due to the painful inflammatory response in the hours following a fracture (Chapter 3 [\[2\]](#)) and high prevalence of this injury. In cage-free aviaries, most of the time in pain at the hurtful (83%) and annoying (64%) categories are also endured as a result of fractures due to the chronic, long-standing character of the pain as a result of the relatively high prevalence of cases of malunion or nonunion/delayed union of the bone (Chapter 3 [\[2\]](#)).

Keel fractures are also a prevalent and important source of pain in conventional cages, but the presence of other (even more prevalent) sources of disabling pain in these systems reduce their relative contribution. In conventional cages, a large proportion of the time spent in disabling pain (of a psychological nature) is associated with the deprivation of a nest. Empirical evidence is well established in showing that the impossibility to find a suitable nest in the moments prior to oviposition, particularly as the sitting phase approaches, is highly disruptive for the hen (Chapter 6 [\[6\]](#)). At this time, hens are willing to pay very high prices for access to a nest site, exposing themselves to situations of danger or choosing to exert great effort (e.g., similar to that exerted to access food after being food-deprived for 28 hours) in exchange for a nest (Chapter 6 [\[6\]](#)). Although we conservatively assumed that the frustration associated with the lack of a nest is temporally restricted to the timing of nesting motivation (just prior to and during oviposition), the fact that hens experience it every day during the laying cycle naturally added up to long time in disabling pain. Some time in pain as a result of nest deprivation is also present in furnished cages and cage-free aviaries (though with a much lower prevalence and relative importance), but in these systems it emerges due to nest competition and the fact that some birds do not find the commercial nest boxes suitable (Chapter 6 [\[6\]](#)).

The frustration emerging from the thwarted possibility to forage and explore the environment similarly accounts for a large fraction of the pain endured at the hurtful and annoying categories in both caged environments ([Figure 9.1](#)), due to the large time budget normally dedicated to these activities and the multiple lines of evidence (Chapter 6 [\[6\]](#)) indicating that the deprivation

of the possibility to forage and explore the environment is very detrimental to the well-being of layers (e.g. high and inelastic demand for foraging opportunities, observations that hens are willing to pay a price in terms of physical pain to forage and fulfil this seeking drive). Likewise, the inability to move around, perform comforting behaviours and adopt typical postures due to the lack of space and restriction of movements, and the ensuing permanent feeling of discomfort (Chapter 6 [6]), accounts for a large proportion of the time spent in annoying pain in conventional and furnished cages.

In all systems, moments of extreme suffering (pain at the excruciating level) are predominantly associated with those injuries and diseases progressing to severe sepsis: acute bacterial peritonitis (a common and painful disorder and the leading cause of mortality in laying hens; Chapter 5 [5]) and vent wounds that become infected (Chapter 4 [3]). To our surprise, the time in excruciating pain due to fatal cases of bacterial peritonitis was estimated to be higher than that due to fatal cases of vent cannibalism (as depicted in the larger size of the purple bars relative to the red bars in Figure 9.1). Although the time in excruciating pain is similar for a hen affected by either condition (1 to 3 hours in both cases; see also <https://welfarefootprint.org/research-projects/laying-hens>), the burden of acute peritonitis is higher due to its higher estimated prevalence (Chapter 8 [8]).

Similarly, the chronic nature of the pain in individuals that survive acute episodes of peritonitis (light purple shades) led to a relatively higher burden of pain at the disabling and hurtful categories than that emerging from pecking wounds (red shades). In fact, the relative time in pain at all intensity categories as a result of pecking injuries (skin wounds, vent wounds, cannibalism) was found to be substantially lower than that of the other conditions studied, particularly those associated with chronic pain. If on the one hand future adjustments in the estimates of the burden of pecking injuries are likely (and desirable) given the scarcity of data on the parameters informing this estimate (Chapter 4 [3] and Chapter 8 [8]), on the other hand we find it difficult that future adjustments will change this finding. This is because the physical pain associated with pecking injuries is predominantly restricted to the acute inflammatory response following tissue damage, with swollen tissues pushing against nerve endings and proinflammatory cytokines acting directly on nociceptors, inducing their persistent sensitization (Chapter 4 [3]). However, even in those cases where inflammation (hence pain) persists for weeks due to a poor wound environment, healing times are relatively short compared to the longer

A blueprint for the comparative analysis of pain

periods of chronic pain associated with keel fractures that do not heal properly, chronic cases of peritonitis or even with the chronic frustration emerging from the deprivation of behaviors hens are highly motivated to perform. Additionally, the proportion of birds affected by pecking injuries is lower (up to 30% in the case of non-fatal injuries, depending on the injury and housing system; Chapter 8 [8]) than the proportion of birds sustaining fractures or, in the case of cages, enduring the deprivations associated with confinement. As a consequence, the burden of physical pain stemming from pecking injuries is bound to be low compared to that of conditions associated with chronic pain¹.

Finally, a relatively high share of the time in disabling pain emerged as a result of depopulation and transport to the slaughterhouse (grey shades in [Figure 9.1](#)). A large burden of pain due to these procedures was expected due to the many concurrent welfare challenges they impose (Chapter 7 [4]), which are in turn exacerbated by the poor health, plumage condition and skeletal strength that is typical of end-of-lay hens, along with the lack of incentives for their careful handling given the low commercial value of their meat. This result achieves a greater significance considering that the pain endured is concentrated on a single day, with the possible multiplicative effects (not modelled here) of multiple sources of uninterrupted pain and distress, making this process a hotspot of welfare loss in the life of these animals.

Absolute burden of pain in different housing systems

[Figure 9.2](#) shows the estimated number of hours endured at the four categories of pain intensity by the average member of a flock in the three housing systems as a result of all welfare challenges combined, whereas in [Figure 9.3](#) the same data is disaggregated by challenge.

¹ Here we address the direct consequence of injurious pecking. Other indirect effects (Figure 4.3 in Chapter 4 [3]) are not considered.

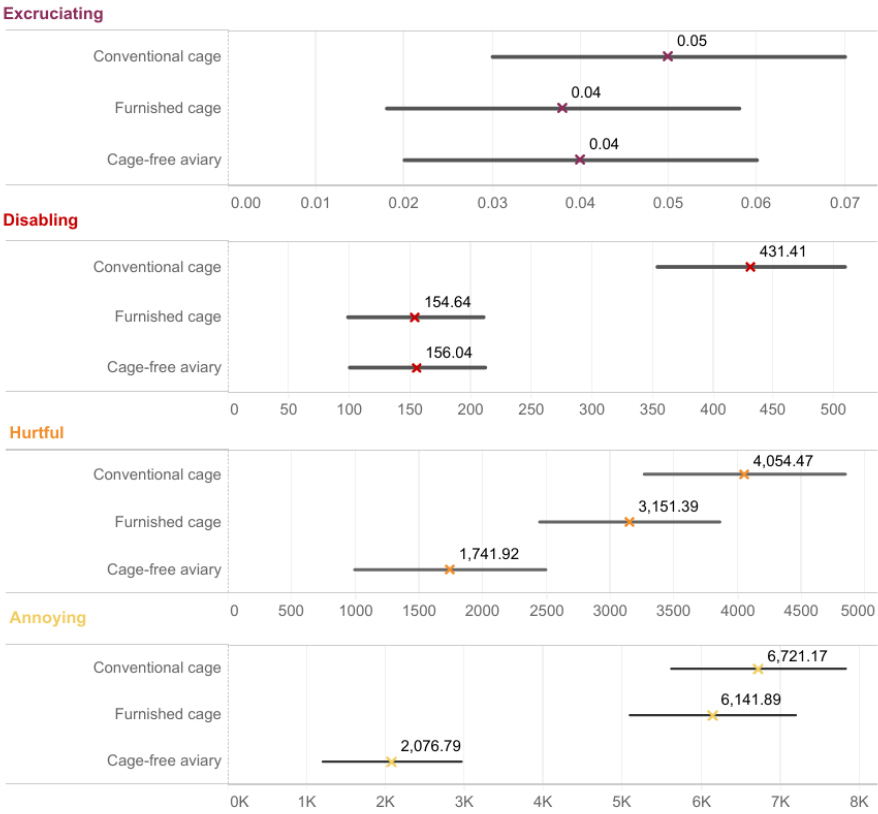


Figure 9.2. Expected time in pain (hours) endured by the average flock member in each housing system and category of pain intensity. Time scales are different for each category of intensity for greater clarity. Only those hours when the bird is awake (16 hours/day) are considered. Error bars represent 95% confidence intervals. An interactive version of the figure is available at <https://welfarefootprint.org/research-projects/laying-hens>.

Through its effects on the welfare challenges investigated here, the transition from conventional cages to cage-free (indoor) aviaries is expected to prevent an average of 275 hours of disabling pain, 2,313 hours of hurtful pain and 4,645 hours of annoying pain for each laying hen that is raised in this system instead of a cage (a 64%, 57% and 69% decrease, respectively, relative to the time in pain in conventional cages). Even if not encompassing all welfare challenges experienced by laying hens, these estimates represent a substantial fraction of the laying cycle (considering an average cycle of 70 weeks, and the onset of

A blueprint for the comparative analysis of pain

lay at 20 weeks, there are approximately 5,600 daylight hours during this period).

Because in conventional cages virtually all aspects of the behavior of laying hens are thwarted, the longer time in pain in conventional cages stems predominantly from the deprivation of behaviors hens are highly motivated to perform and the discomfort associated with the lack of space and the restriction of movement (Chapter 6 [6]). Even though the time in pain as a result of keel bone fractures is higher in aviaries ² (larger orange bars in Figure 9.3, corresponding to 40, 545 and 507 hours more in disabling, hurtful and annoying pain, per hen, in aviaries as a result of keel fractures), the difference is much lower than that needed to compensate for the chronic psychological pain emerging from the frustration of important behavioral needs in cages.



² The expectation that the time in pain due to keel fractures is higher in cage-free settings is greatly affected by the extent to which the prevalence of keel fractures are higher in cage-free settings. As discussed in Chapter 8 [8], we used prevalence values as reported in published research. However, a recent study suggests that the prevalence of fractures is likely to be greatly underestimated in conventional cage systems [14]. Another recently published review on the prevalence of keel fracture in laying hens also concludes that the link between housing system and keel bone fracture prevalence may not be supported [8].

Figure 9.3. Expected time in pain (hours) endured by the average flock member in each housing system and category of pain intensity, disaggregated by harm. Time scales are different for each category of intensity for greater clarity. Only those hours when the bird is awake (16 hours/day) are considered. An interactive version of the figure is available at <https://welfarefootprint.org/research-projects/laying-hens>.

Although the comparison involves welfare challenges of different nature (physical and psychological), it is important to remember that the criteria used for the attribution of intensity in both cases is the same, focusing on (1) the importance of the pain signal to promote adaptive behaviors and (2) the disruptive character of the pain experience (Chapter 1 [1]). For example, pain at the disabling level is characterized by its power to overrule most other bids for behavioral execution, with continuous disruptive symptoms that prevent all forms of positive welfare. Such a criterion matches closely the frustration of a hen that cannot find a nest in the moments prior to oviposition (Chapter 6 [6]). At this time, not only is all attention diverted to this seeking drive, but hens are also willing to pay very high prices (in terms of effort and exposure to danger) for access to a nest site, similar to those paid to access food following long periods of food deprivation. Such a degree of disruption, along with the evolutionary value of nests, indicate that the frustration emerging from lack of a suitable site just prior to oviposition is most likely of a disabling nature (see Box 1.2 in Chapter 1 [1] for definitions). A similar rationale applies to the attribution of disabling pain in the hours, and day, following a keel fracture (see Chapter 3 [2]). The experience of intense pain as a result of a fracture is extremely adaptive, as it ensures the reduction of movement needed to facilitate healing and protect the individual from further injury, as well as the presumably higher need for rest at a time of increased metabolic demand on the body. The adaptive value of intense pain at this stage is reflected at the physiological and anatomical levels. The periosteum has an extremely dense sensory and sympathetic innervation, where most of the mechanosensory fibers activated immediately after fracture are believed to be located. After the injury, an influx of inflammatory cells to the fracture site results in the release of inflammatory mediators that will cause the swelling of the site (hence pain through activation of pressor receptors) and activate undamaged nociceptors [11–13]. Sensitization of the peripheral and central nervous systems may also occur, manifesting as increased sensitivity to pain [12,13]. Accordingly, existing data on the behavior of birds (pigeons) in the immediate days following a fracture reveal substantial changes in behavior and alertness, even among individuals receiving high doses of analgesia - all patterns compatible with the definition of disabling pain.

A blueprint for the comparative analysis of pain

The average time spent in excruciating pain is similar in all systems, resulting mostly from fatal events that affect only a small fraction of individuals (e.g. the proportion of the flock estimated to die from acute cases of egg peritonitis syndrome is expected to range from 0.6 to 3.2%; Chapter 8 [8]). Although only a few minutes of excruciating pain are expected for the average population member when the time in pain is weighted by the prevalence of these events, the time in excruciating pain endured by the individuals affected is, however, very high (2.25 hours on average, mostly due to severe sepsis). Considering a flock of 50,000 birds, approximately 2,000 hours of excruciating pain per flock are expected as a result of the conditions analysed here.

The estimates also point to a higher burden of pain in furnished cages compared to cage-free aviaries, though less pronounced than in the comparison with conventional cages due to the opportunity to partially fulfil some of the behavioral motivations in the furnished system (chiefly the need to lay an egg in a nest). Hens from furnished cages and aviaries are expected to spend a similar time in disabling and excruciating pain (approximately 154-156 hours and 2.5 minutes per flock member, respectively), but on average 1,410 hours of hurtful pain, and 4,065 hours of annoying pain are estimated to be averted during the laying cycle for each laying hen that is raised in an aviary instead of a furnished cage.

As expected, furnished cages are better from a welfare perspective than conventional cages. Despite the relatively similar times in annoying pain, laying hens will spend somewhat shorter times in hurtful pain (~25% decrease), and substantially shorter times (~64% decrease) in disabling pain in furnished than in conventional cages (on average 277 hours of disabling pain are averted per bird).

It has been often highlighted that each housing system has unique welfare challenges [14–16], and that no system is ideal for the welfare of laying hens. While our findings are in line with these observations, these remarks should not be taken to imply that different systems are equivalent in terms of their overall impact on welfare. The possibility to quantify, and compare, the impact of major welfare challenges on the burden of pain endured by commercial layers strongly indicate that cage-free aviaries are preferred both to conventional cages and furnished cages in efforts to reduce farm animal suffering.

The empirical estimates presented here are conservative, given our deliberate attempts to avoid inflating the welfare benefits of cage-free housing based on our prior beliefs and conceptions, particularly with regard to the estimates of

the prevalence of the welfare challenges. As discussed, estimates of the prevalence of different challenges are most likely to reflect a situation where the maturity of cage-free housing is not yet on par with that already achieved by caged systems. For instance, based on data reported since the year 2000, the proportion of hens with pecking wounds and the proportion of deaths due to cannibalism were assumed to be higher in cage-free systems, even though there is evidence to indicate that tissue pecking (linked to mortality) is also reduced over time in cage-free facilities [17]. Indeed, our investigation of plumage damage scores suggested their improvement over the years (Chapter 8 [8]). Likewise, in a recent review the authors conclude that the expectation that keel bone fractures are more prevalent in cage-free systems may not be supported [18]. Still, due to the scarcity of data to analyse formally the influence of these confounders, the prevalence of keel fractures was assumed to be higher in cage-free facilities (represented by the range of prevalence values reported in commercial settings). Additionally, to avoid inflating the psychological pain endured by caged hens, in those cases where evidence was conflicting as to the importance of some behaviors and resources for hens (e.g. use of perches during the day), we conservatively assumed that their absence in conventional cages was not a major driver of psychological pain (Chapter 6 [6]). Overall, this makes us confident on the robustness of the effects reported.

Important Considerations

The estimates presented here do not emerge from an exhaustive analysis of every possible welfare challenge affecting commercial layers, focusing instead on key challenges experienced during the laying cycle only. Here we examine the influence of aspects not factored in the quantitative analysis previously presented. Most of these factors are expected to increase further the burden of pain in cages relative to cage-free housing (e.g. longer laying cycles, induced molting, higher levels of fear, a perceived lack of control), although some (e.g. foot lesions, air quality) possibly have the opposite effect.

In the latter category, one such problem is footpad dermatitis, an inflammation of the plantar region of the foot that can progress to ulceration and the development of necrotic lesions. If lesions are infected (most commonly by *Staphylococcus*) they can ultimately lead to a painful condition known as bumblefoot (Figure 9.4). Laying hens are possibly at a higher risk of footpad lesions in non-cage systems, as one of the main risk factors for its

A blueprint for the comparative analysis of pain

development is the presence of wet litter. Footpad dermatitis has received considerable attention in broilers (where it is more common, as broilers spend their whole life in close contact with litter mixed with droppings), but the extent to which the problem affects commercial layers in different systems is not entirely clear (e.g., while some studies found a higher prevalence in cage-free settings, others have failed to find differences [16,19,20]). Likewise, the reporting of the relatively low incidence of the more severe and painful manifestations in layers [16,21–23] makes it unlikely that consideration of this harm would affect the estimates substantially to the point of changing any of the conclusions.

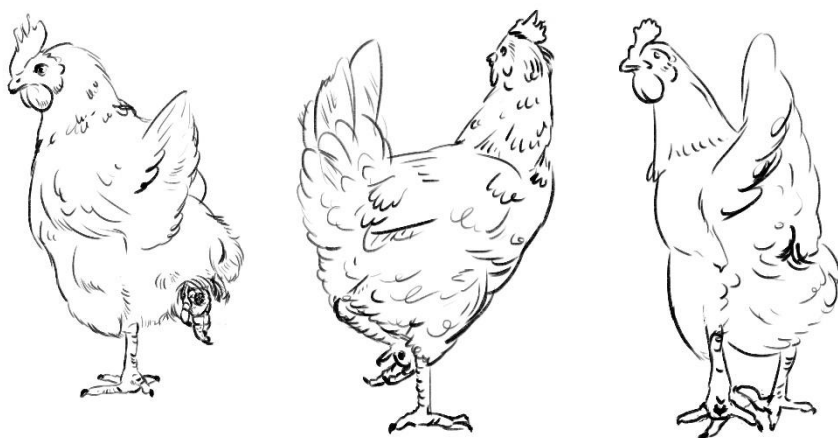


Figure 9.4. Laying hens affected by footpad dermatitis. Illustration: Olga Kurkina.

Another common concern involving cage-free facilities is the quality of the air, particularly with regard to the concentration of ammonia (NH_3). As NH_3 originates from the decomposition of nitrogen-containing manure, its concentration can be generally higher in litter-based housing compared to caged systems with manure belts [24]. There is evidence to indicate that ammonia is aversive to chickens (eg. when given a choice between 4, 11, 20, and 37 ppm of atmospheric ammonia, broilers avoid the higher concentrations [25]), so it is not unreasonable to suppose that through the potentially detrimental effect on the respiratory system [24] and on mucous membranes, high concentrations of ammonia can lead to a prolonged state of

discomfort [26]. This is an important welfare concern, which is likely to increase the estimated time in pain endured in cage-free facilities, depending on the prevalence of (cage and cage-free) facilities where manure is not regularly removed and the ventilation flow is insufficient, and on the time endured in discomfort (e.g. in temperate regions, higher levels of ammonia in cage-free relative to cage housing have been found in winter, but not in summer [27,28]).

While one should not be complacent about these challenges (and seek solutions to their mitigation), the evidence reviewed in this book strongly indicates that it is unlikely that consideration of the time in pain they impose would exceed that resulting from the sustained states of aversiveness associated with the many restrictions in the caged environment. In fact, several factors are expected to potentiate the welfare challenges in cages. One such factor is the perception of lack of control in caged environments (Box 9.1). A sense of control over adverse conditions has been shown to be one of the most critical components of mental well-being, both in humans and animals [29–31]. Animals unable to escape aversive conditions are known to develop severe emotional distress in the form of helplessness [29], a depressive-like state recognized as an indicator of severe suffering. Laying hens housed in conventional cages are virtually prohibited from expressing any form of agency [32], highly motivated behaviors, to distance themselves from cage mates, resolve social conflicts or even adopt typical postures and movements needed for a minimum level of comfort and proper thermoregulation - all conditions conducive to the development of states akin to learned helplessness. Often, caged hens are also kept in conditions of extremely low light intensity, compounding this effect. Conversely, by offering birds with novel situations on a daily basis, with freedom of movement and a larger space to explore, cage-free environments can provide hens with the perception of control over their decisions and surroundings [33], making the emergence of these negative mental states improbable.

Box 9.1 Lack of Control and Helplessness in Caged Housing

The very reason for the existence of pain, pleasure and other affective states is to enable organisms to act in an adaptive manner, i.e. to exert control over their fate [34]. However, in captive environments the extent to which individuals can control what happens to them is severely impaired [35]. What happens when animals are entirely denied the possibility to pursue a strong motivation or pleasant activity, or to exert any control over a threatening or painful situation for a prolonged period of time?

Research in both animals and humans has shown that a sense of control over adverse conditions is one of the most critical components of mental well-being [29,34]. Animals that perceive to have control over their fates have been shown to benefit from several robust effects, such as improved coping with stress and a greater ability to function even in potentially dangerous situations [34]. In chickens, there is wide evidence (Chapter 6 [6]) that expressions of agency, whereby the animal engages actively with the environment [36], are rewarding, independently of their outcome. Typical cases are observations of contrafreeloading (in which individuals consistently work to access a reward even when it is also freely available) and high levels of motivation to seek opportunities to explore novel objects of zero utility [35,37–42].

Conversely, animals unable to interact with their surroundings and escape aversive conditions have been shown to develop severe emotional distress in the form of helplessness [29], a debilitating emotional state whereby individuals become passive, lose motivation, the ability to experience joy or pleasure, and have difficulty learning that their responses are consequential, even in subsequent situations where choice is available. Not surprisingly, the development of learned helplessness in animals has been long used as a model for clinical depression in humans [43,44]. Depression-like states such as learned helplessness are in fact recognized as indicators of severe suffering, to the point that experiments inducing learned helplessness have been banned in some countries [45].

A high degree of passivity, suggestive of learned helplessness, has been demonstrated in chickens exposed to inescapable electric shocks [46]. After frustrated attempts to escape the birds give up, indicating that chickens, too, can experience such a depressive-like state. Laying hens housed in barren cage systems are virtually prohibited from expressing any form agency [32] or express any motivated behavior. Furthermore, in the crowded cage conditions hens appear to be in a chronic state of social

stress, unable to distance themselves from cage mates, express dominance relations or resolve social conflict by means of aggression [47] - all conditions conducive to the development of states akin to learned helplessness. Cage-free environments, in turn, offer birds with a larger space to explore and a number of novel situations on a daily basis. Perhaps more important, freedom of movement is likely to provide the animals with the perception of control over their surroundings and decisions [33]. Within the limits of the aviary, hens can choose what to do, where and for how long. They can hide or distance themselves from conspecifics they would rather avoid, making the emergence of mental states associated with a perceived lack of control much more unlikely.

A perceived lack of control can also potentiate fear [48]. Fear can be extremely damaging if intense or persistent, or if the animal is prevented from responding to the threat [49], being demonstrably linked to stress [50,51]. While in the complex aviary environment birds are likely to be exposed to a higher degree of mild intermittent stress (e.g. constant exposure to unfamiliar flockmates), confinement in the barren cage environment can generate a more sustained state of fear as a result of helplessness and deprivation [52]. As outlined in greater detail in [Box 9.2](#), prior research indicates that, where differences are present, laying hens are more fearful when confined in cages (a higher level of fearfulness has also been found in broiler chickens and common pheasant hens housed in conventional cages compared to enriched and cage-free systems [53,54]). Should the time in pain due to fearfulness be considered, it would increase the burden of pain in the caged conditions.

Additionally, in cages (particularly conventional cages) birds are also deprived of the opportunity to enjoy positive affective experiences, with no possibility of distraction or compensation for the negative experiences endured ([Box 9.3](#)). If it was possible to objectively determine the extent to which positive affective experiences can compensate for time in pain, estimates of the time spent in pain in cage-free environments would be reduced further relative to that experienced in cages.

Box 9.2 Differences in fearfulness in cage and non-cage systems

Fear is an adaptive response that helps individuals react appropriately when faced with threats to their physical integrity and/or reproductive capability. It is a hardwired and endogenously coordinated function that responds to unconditional threatening stimuli, and with time, to other stimuli individuals learn to fear [55]. Yet despite its adaptive nature, fear can be a powerful and damaging stressor if intense or persistent, or if the animal is prevented from properly responding to the threatening situation [49]. Fear has been demonstrably linked to stress [50,51], potentially providing a more reliable indicator of how well the animal is coping with the environment [56] than traditional endocrine and immunological markers (the interpretation of endocrine and immunological indicators is complex, non-specific and confounded by many factors [57]). Furthermore, because it inhibits other motivational systems, fear impairs the ability of birds to express motivated behaviors, interact socially and adapt to change.

Several measures of fearfulness have been devised, such as the response of birds to unfamiliar objects, to novel environments and the time spent in tonic immobility (TI) as a result of physical restraint (Box 4.1 in Chapter 4 [3]). In general, the higher the level of fearfulness, the longer individuals stay immobile. Where differences were observed, laying hens were shown to be more fearful when confined in cages (conventional or furnished) than when kept in cage-free housing, with significantly longer TI durations and less inductions needed to induce tonic immobility [16,58–60]. This is despite the fact that TI tests may have low sensitivity to environmental differences [61], and might in fact induce more fear in non-cage hens because of the greater difficulty (hence stress) to catch them. Caged hens have also shown more panic reactions to novel objects, similarly indicating a higher level of fearfulness [62].

The housing environment during the rearing period (prior to lay) has also been linked to long-term differences in fearfulness levels. For example, birds reared in a complex aviary system have been shown to be less fearful as adults than those reared in a simpler cage environment ([63,64], but see [65]), a result in line with evidence showing that early exposure to greater environmental complexity can act as a buffer against stress in the future [66]. There is also some evidence that the fearfulness of hens in cages, as measured by the TI test, increases considerably with time: in one study, differences in TI response between caged and cage-free hens were substantial at 70 weeks of age, but not at 30 weeks [60].

Box 9.3 Lack of positive affective experiences in cages

Welfare is a multidimensional concept, comprising the many dimensions of an individual's physical and psychological states. These states can be associated with both positive and negative hedonic experiences of varying intensities [67]; at any point, the subjective well-being of individuals will depend on a balance between positive and negative states.

In this book our focus has been on the measurement of affective states of negative valence, namely those related to the presence of what we operationally classified into physical and psychological pain. This perspective is at the core of most research on the valuation of human health and welfare, and forms the basis of various policies guiding efforts at poverty alleviation and disease mitigation. For example, the measurement of negative states is also the method of choice in epidemiology and public health research to identify and mitigate sources of loss of quality of life in human populations [68]. Additionally, the mitigation of some sources of psychological pain (e.g. the frustration emerging from the deprivation of highly motivated behavior) necessarily translates into the provisioning of positive affective experiences.

Still, should it be possible to objectively determine the extent to which positive affective experiences can compensate for time in pain, this would allow for a number of refinements. For instance, many would accept that the pleasure experienced by laying hens given access outdoors would compensate (but see [69]) for the potential additional pain from injuries in a fraction of animals due to accidents in this less controlled environment. Using a human parallel, this would equate to the higher well-being reported among inmates allowed time playing and socializing in the prison yard, even though it involves a higher number of painful injuries due to fights and occasional accidents. Additionally, access to experiences of positive valence can have effects that go beyond compensating for pain. In laying hens, evidence suggests that the provision of positive experiences (such as enrichment) can directly improve their affective state and make hens more resilient to aversive stressors: hens in enriched environments appraise aversive stimuli less negatively and mount smaller stress responses to restraint [70].

Although knowledge for the objective quantification of the flow-through effects of a life completely devoid of positive affect is not yet available, this factor should not be left aside in decision contexts where the welfare of animals is at stake [71–74]. In cage systems, particularly conventional cages, birds are deprived of opportunities to enjoy positive experiences. This effectively translates into no

A blueprint for the comparative analysis of pain

possibility of compensation even for the mildest forms of suffering. Naturally, the goal of ensuring a life 'worth living' for farm animals, namely a life where the balance of experiences throughout life is positive [73,74], cannot be achieved in these systems.

The possibility to fulfil various behavioral needs in aviaries is likely to have other positive flow-through effects, not only in terms of an improved sense of general well-being emerging from positive experiences but, importantly, in their capability to mitigate the unpleasantness of painful experiences [34]. In chickens, physical pain has been shown to be partially or completely suppressed by motivationally induced changes in attention [75–78], possibly through an opioid-mediated mechanism of endogenous analgesia. In general, the higher the motivation to engage in a behavior (hence attention diverted to it), the higher the degree of analgesia, with some evidence that inflammation subsides too [76,79]. These findings suggest that in more complex environments the salience of pain (and possibly the duration of the healing process) is reduced relative to that experienced in barren conditions [80] (Box 9.4). In our analysis, we conservatively assumed that the pain endured by an individual affected by a fracture, or wound, or episode of peritonitis, was the same in all housing systems. However, should this possibility hold, it could translate into hundreds of hours more (in addition to those estimated) of hurtful and annoying pain averted in cage-free aviaries compared to cage systems.

Box 9.4 Partial suppression of pain in cage-free environments?

Lack of control and fear can exacerbate not only the effects of aversive conditions, reducing the individual's ability to cope with negative affective states, but also the very perception of physical pain. For example, telling people they can give a signal to stop a medical procedure has been shown to effectively reduce their pain, even if they never use the signal [48]. Similarly, human patients who express greater fear of pain are more likely to focus on their painful experiences [81]. Furthermore, increased pain sensitivity is seen when people expect pain [82]. In general, the more attention diverted to pain, the greater its perceived intensity. Conversely, distraction is one of the most commonly used methods to control pain in human patients [83], particularly in poor settings where the cost of painkillers is prohibitive [84].

In chickens, physical pain has been similarly shown to be partially or completely suppressed by motivationally induced changes in attention. In a series of experiments, Gentle and colleagues [75–77] used a model of a tonically painful inflammation akin to that experienced in acute gouty arthritis [78], a highly painful condition resulting from the formation of sodium urate crystals in the joints. In these experiments, pain was induced by an intra-articular injection of sodium urate into the chicken's ankle, lasting for at least three hours, during which the chicken showed typical pain-related behaviours, including sitting, limping, and an unwillingness to walk or stand on the affected ankle [85]. However, when urate-injected birds were provided with exploration and social opportunities [75], or placed in situations designed to increase their feeding motivation [77], complete analgesia or significantly less pain-related behaviours was observed. For example, in enriched environments, birds showed no signs of pain, fear or distress, behaving instead as normal, alert birds whose attention was directed towards exploring the environment. The possibility that the absence of pain-related behaviors indicate the absence of pain was further confirmed by additional experiments, in which urate-injected birds were treated with naloxone, a medication that blocks the effects of opioids [77]. For these birds, pain-related behaviors were not suppressed, suggesting a role of endogenous opioids in the analgesic effects observed. Overall, these results indicate that the possibility to fulfil any strong motivation results in endogenous analgesia, presumably by diverting attention away from the pain [77,85]. The authors suggest that the higher the motivation to engage in a behavior (hence attention diverted to it), the higher the degree of endogenous analgesia: when the birds could engage in preferred behaviors, such as nesting and feeding (following a period of food deprivation), complete endogenous analgesia was observed [77]. Additionally, not only was pain

suppressed, but also the clinical signs of inflammation, presumably through the reduced effect of the neurogenic component of inflammation [76,79].

These findings have important implications for welfare comparisons among animals kept in conditions offering different possibilities of distraction. They suggest that in more complex environments the salience of pain (and possibly healing times) is reduced relative to that in barren conditions [80]. It follows that pain resulting from harms such as bone fractures, foot problems and peritonitis is possibly reduced in cage-free housing. This possibility should be investigated further.

A further increase in the burden of pain in cages relative to aviaries is also expected when one considers that caged hens are often depopulated at a later age (in the present analysis, we considered the same laying cycle, of about 70 weeks, for all housing systems). The inclusion of the last weeks of life of a caged hen in the analysis would possibly increase the burden of pain in cages substantially. First, by extending by many weeks the chronic states of pain associated with the welfare challenges analysed. Second, by including the effect of procedures such as induced molting, a practice associated with important welfare concerns and still widely adopted outside Europe. (for example, in the United States, 60-80% of commercial caged layers are estimated to be molted [86–88], compared to less than 15% in cage-free production [86]). Finally, by encompassing the multiple comorbidities present in the last weeks of life in caged hens [35]. Again, survey data from the United States indicate that cage-free flocks are depopulated when mortality reaches on average 6.4%, compared to 10.5% for flocks raised in conventional cages [86]. We hypothesize that caged systems can extend poor health states for a longer period of time [9], as health conditions producing weakness, lethargy, or reduced movement can be prolonged since no displacement or effort is needed for sick or critically ill individuals to keep eating and drinking, allowing for a higher prevalence of morbid individuals still in production.

Going Forward

Although most of the egg laying hens alive are still housed in conventional cages (Chapter 2 [7]), large-scale changes towards cage-free production are underway in many countries. Here we quantified the impact of this transition, estimated based on the time at each intensity of pain prevented for each hen no longer raised in a cage. To this end, we focused on key welfare challenges to which laying hens experience during the laying cycle.

The present estimates point to thousands of hours of pain prevented for every laying hen raised in an indoor cage-free aviary instead of a cage (both conventional and furnished). Even when considering all the uncertainty in the parameters used in the calculations (intensity and duration of exposure to harms, their occurrences and prevalence in the population), the risk of these claims not holding is low. On the contrary, we have argued that the expansion of the analysis towards the inclusion of other challenges not presently considered will possibly amplify the differences found.

While our estimates indicate that cumulative time in pain endured in cage-free aviaries is substantially lower than in caged conditions, they also show that a long time in pain is still spent in these environments, predominantly as a result of ailments associated with selection for high levels of egg production. The high prevalence of painful reproductive disorders (such as egg peritonitis syndrome) is an illustrative example, along with the tendency to perform injurious pecking, a behavior that *'seems to have been selected inadvertently as a trait that hitchhiked on the back of selection for early onset of egg laying and the ability to use calcium to produce good-quality eggshells'* [39]. Even more importantly, selective breeding for productivity and an extended laying period, and the resulting use of structural bone for egg shell production, has led to the widespread occurrence of osteoporosis and increased bone fragility [89]. Bone fractures were, by far, the greatest contributor to the burden of pain endured by laying hens, accounting for 66%, 82% and 64% of the hours in disabling, hurtful and annoying pain in cage-free aviaries (before depopulation). The continued selection for persistence of lay production, with expected cycle increases to 100-120 weeks, is likely to result in birds that are even more susceptible to osteoporosis, not only increasing the time spent in pain, but also the pain endured per unit of time.

Improvements in these conditions will likely require the use of genetic selection techniques (e.g., both bone strength characteristics and injurious

pecking tendencies have been found to respond readily to selection [14,90]), even though this may be in detriment to laying performance [91]. The prioritization of traits other than production would also have multiple additional welfare benefits, reducing the incidence of production diseases such as egg peritonitis syndrome, the frequency of vent trauma (that often triggers episodes of vent cannibalism) and potentially redirecting energy for the proper expression of behavioral traits (genetic breeds selected for high egg production in a small cage without enrichment and without the requirement of well-developed motoric, sensory, or cognitive abilities is thought to have caused undesirable side effects, including more aggressive social behaviour [35] and a reduction of brain size with possible functional consequences [14]). Also, at a time of global sanitary crisis, genetic selection that does not prioritize a single trait (productivity), but maintains a reasonable level of rusticity and balance between the different needs of the organism (including a healthy immune system), should contribute to reduce the likelihood of future infectious disease outbreaks that spill over to the human population [92–94].

The move away from caged production provides an opportunity for the selection of birds sufficiently robust to withstand the more complex and varied social and environmental conditions outside the cages [91]. However, policies and reforms that compromise productivity will meet with resistance [95]. In this sense, some authors have proposed the development of mechanisms to raise incentives for a low prevalence of fractures and injuries in laying hens, coupled with penalties for a high prevalence [96]. The monitoring of these and other conditions could be greatly improved by the enforcement of rigorous pass/fail criteria in inspections at the abattoir, by independent welfare auditing at depopulation and the implementation of mechanisms to improve transparency in the production and finishing chains [35].

Research Possibilities

Surprisingly little research has been dedicated to the understanding of the impact of different welfare challenges at the individual level (where suffering occurs). Little is known about clinical evolution of the various welfare issues affecting commercial layers (e.g. healing times, duration of exposure), the likelihood of different clinical outcomes, rates of recurrence, and how adversely welfare harms are perceived by the individuals affected. Similarly, knowledge on case-fatality rates, comorbidity patterns and the prevalence of different conditions over the laying cycle is scant.

Although the existing lack of knowledge and data has been incorporated into the uncertainty intervals associated with the parameters used, there is much room for improving the estimates presented here as knowledge advances - we very much welcome any contribution that allows adjusting and updating this analysis. Expansion and refinement is not only possible given the modular structure of the analytical framework used, but desirable.

In previous chapters we highlighted the main research gaps that emerged in the analysis of the individual welfare challenges examined. In addition to these specific data gaps, we believe that longitudinal studies targeting individual birds in commercial conditions are very much needed. They can help understand (1) the extent to which each welfare harm disrupts the routine and behavior of the individuals affected, hence helping establish the degree to which conditions are perceived as aversive; (2) the recurrence of harms (how many wounds, attacks, fractures, or disease episodes an individual animal endures over the laying cycle); (3) the temporal evolution of each harm, including fatal conditions; (4) the rates of spontaneous healing in the case of injuries and diseases; (5) patterns of comorbidity and (6) the extent to which behavioral opportunities are realized in cage-free settings, and how this changes over time. The availability of methods to tag and video-record animals should greatly facilitate efforts in this direction.

Similarly, comprehensive necropsy studies based on the random and systematic sampling of apparently healthy hens (those still alive until depopulation) kept in different production contexts can help determine the prevalence of a number of non-fatal conditions in layer flocks, and the extent to which these conditions are correlated. Data collected in this way can shed light on potentially neglected sources of suffering in layer flocks, and on the risk factors for the development of important conditions (such as keel fractures). In fact, necropsy surveillance is already performed routinely in many countries, but concentrated on indicators related to food safety. With

A blueprint for the comparative analysis of pain

protocols that include the collection of additional data, existing surveillance efforts could provide valuable welfare information.

Finally, the possibility that in environments where animals can fulfil important behavioral needs the salience of pain and healing times are reduced relative to that experienced in barren and confined conditions should be investigated further. Should it be supported, it would have important implications for policies and reforms in all areas of animal welfare.

References

1. Alonso WJ, Schuck-Paim C. [The Comparative Measurement of Welfare in Animals: Cumulative Time in Pain as a Universal Metric with Biological Meaning](#). In: Schuck-Paim C, Alonso WJ, editors. *Quantifying Pain in Laying Hens*. Independently published. 2021.
2. Schuck-Paim C, Alonso WJ. [Quantifying the Pain due to Keel Bone Fractures in Laying Hens](#). In: Schuck-Paim C, Alonso WJ, editors. *Quantifying Pain in Laying Hens*. Independently published. 2021.
3. Schuck-Paim C, Alonso WJ. [Welfare Implications of Injurious Pecking in Laying Hens](#). In: Schuck-Paim C, Alonso WJ, editors. *Quantifying Pain in Laying Hens*. Independently published. 2021.
4. Schuck-Paim C, Alonso WJ. [The Last Day of a Hen's Life: Depopulation and Transport](#). In: Schuck-Paim C, Alonso WJ, editors. *Quantifying Pain in Laying Hens*. Independently published. 2021.
5. Schuck-Paim C, Alonso WJ. [Egg Peritonitis Syndrome: A Painful and Prevalent Disease in Commercial Layers](#). In: Schuck-Paim C, Alonso WJ, editors. *Quantifying Pain in Laying Hens*. Independently published. 2021.
6. Schuck-Paim C, Alonso WJ. [The Burden of Psychological Pain in Laying Hens: Behavioral Deprivation](#). In: Schuck-Paim C, Alonso WJ, editors. *Quantifying Pain in Laying Hens*. Independently published. 2021.
7. Schuck-Paim C, Alonso WJ. [The Life of Commercial Laying Hens](#). In: Schuck-Paim C, Alonso WJ, editors. *Quantifying Pain in Laying Hens*. Independently published. 2021.
8. Schuck-Paim C, Alonso WJ. [The Prevalence of Welfare Harms Affecting Commercial Layers in Different Housing Systems](#). In: Schuck-Paim C, Alonso WJ, editors. *Quantifying Pain in Laying Hens*. Independently published. 2021.
9. Schuck-Paim C, Negro-Calduch E, Alonso WJ. [Laying hen mortality in different indoor housing systems: a meta-analysis of data from commercial farms in 16 countries](#). *Sci Rep*. 2021;11: 3052.
10. Cotra A. [How will hen welfare be impacted by the transition to cage-free housing?](#) Open Philanthropy; 2017.
11. Richards GJ, Nasr MA, Brown SN, Szamocki EMG, Murrell J, Barr F, et al. [Use of radiography to identify keel bone fractures in laying hens and assess healing in live birds](#). *Vet Rec*. 2011;169: 279.
12. Nasr MAF, Nicol CJ, Murrell JC. [Do laying hens with keel bone fractures experience pain?](#) *PLoS One*. 2012;7: e42420.

A blueprint for the comparative analysis of pain

13. Desmarchelier M, Troncy E, Beauchamp G, Paul-Murphy JR, Fitzgerald G, Lair S. [Evaluation of a fracture pain model in domestic pigeons \(*Columba livia*\)](#). *Am J Vet Res*. 2012;73: 353–360.
14. Mehlhorn J, Petow S. [Smaller brains in laying hens: New insights into the influence of pure breeding and housing conditions on brain size and brain composition](#). *Poult Sci*. 2020;99: 3319–3327.
15. Lay DC Jr, Fulton RM, Hester PY, Karcher DM, Kjaer JB, Mench JA, et al. [Hen welfare in different housing systems](#). *Poult Sci*. 2011;90: 278–294.
16. Rodenburg TB, Tuytens FAM, de Reu K, Herman L, Zoons J, Sonck B. [Welfare assessment of laying hens in furnished cages and non-cage systems: an on-farm comparison](#). *Anim Welf*. 2008;17: 363–373.
17. Berg C, Yngvesson J. [The transition from battery cages to loose housing systems and furnished cages for Swedish laying hens](#). In: World Poultry Science Association, editor. *Proceedings of the European Poultry Conference*. 2006.
18. Rufener C, Makagon MM. [Keel bone fractures in laying hens: a systematic review of prevalence across age, housing systems, and strains](#). *J Anim Sci*. 2020;98: S36–S51.
19. Freire R, Cowling A. [The welfare of laying hens in conventional cages and alternative systems: first steps towards a quantitative comparison](#). *Anim Welf*. 2013;22: 57–65.
20. Sherwin CM, Richards GJ, Nicol CJ. [Comparison of the welfare of layer hens in 4 housing systems in the UK](#). *Br Poult Sci*. 2010.
21. van Niekkerk T, Gunnink K, van Reenen K. [Welfare Quality® assessment protocol for laying hens](#). Result assessment for 122 flocks. Report 589. 2012.
22. Heerkens JLT, Delezie E, Rodenburg TB, Kempen I, Zoons J, Ampe B, et al. [Risk factors associated with keel bone and foot pad disorders in laying hens housed in aviary systems](#). *Poult Sci*. 2016;95: 482–488.
23. Heerkens JLT, Delezie E, Ampe B, Rodenburg TB, Tuytens FAM. [Ramps and hybrid effects on keel bone and foot pad disorders in modified aviaries for laying hens](#). *Poult Sci*. 2016;95: 2479–2488.
24. Hofmann T, Schmucker SS, Bessei W, Grashorn M, Stefanski V. [Impact of Housing Environment on the Immune System in Chickens: A Review](#). *Animals (Basel)*. 2020;10.
25. Jones EKM, Wathes CM, Webster AJF. [Avoidance of atmospheric ammonia by domestic fowl and the effect of early experience](#). *Appl Anim Behav Sci*. 2005;90: 293–308.

26. David B, Mejdell C, Michel V, Lund V, Moe RO. [Air Quality in Alternative Housing Systems may have an Impact on Laying Hen Welfare. Part II—Ammonia](#). *Animals (Basel)*. 2015;5: 886–896.
27. Green AR, Wesley I, Trampel DW, Xin H. [Air quality and bird health status in three types of commercial egg layer houses](#). *J Appl Poult Res*. 2009;18: 605–621.
28. Zhao Y, Shepherd TA, Li H, Xin H. [Environmental assessment of three egg production systems—Part I: Monitoring system and indoor air quality](#). *Poult Sci*. 2015;94: 518–533.
29. McMillan FD. [Maximizing quality of life in ill animals](#). *J Am Anim Hosp Assoc*. 2003;39: 227–235.
30. McMillan F. [What Is Distress? A Complex Answer to a Simple Question](#). In: McMillan F, editor. *Mental Health and Well-being in Animals*. CAB International; 2020. pp. 140–155.
31. Weary D. [Suffering, Agency, and the Bayesian Mind](#). In: McMillan F, editor. *Mental Health and Well-being in Animals*. CAB International; 2020. pp. 156–166.
32. Spinka M. [Animal agency, animal awareness and animal welfare](#). *Anim Welf*. 2019;28: 11–20.
33. Tahamtani FM, Hansen TB, Orritt R, Nicol C, Moe RO, Janczak AM. [Does rearing laying hens in aviaries adversely affect long-term welfare following transfer to furnished cages?](#) *PLoS One*. 2014;9: e107357.
34. McMillan FD. [The Mental Health and Well-being Benefits of Personal Control in Animals](#). 2 edition. In: McMillan FD, editor. *Mental Health and Well-Being in Animals (2nd edition)*. 2 edition. CAB International; 2020. pp. 67–81.
35. Grandin T, Johnson C. [Animals Make Us Human: Creating the Best Life for Animals \(Chapter 7: Chickens and other Poultry\)](#). First Edition. Mariner Books; 2010.
36. Spinka M, Wemelsfelder F. [Environmental Challenge and Animal Agency. Animal welfare](#). CABI International; 2011.
37. Gunnarsson S, Matthews LR, Foster TM, Temple W. [The demand for straw and feathers as litter substrates by laying hens](#). *Appl Anim Behav Sci*. 2000;65: 321–330.
38. Matthews LR, Chandler JH. [Assessing behavioural demand functions: issues](#). In: Duncan IJH, Widowski TM, Haley DB, editors. *Proceedings of the 30th International Congress of the International Society for Applied Ethology*. Campbell Centre for the Study of Animal Welfare; 1996. p. 54.

A blueprint for the comparative analysis of pain

39. Nicol C. [The Behavioural Biology of Chickens](#). CAB International; 2015.
40. Dawkins MS, Beardsley T. [Reinforcing properties of access to litter in hens](#). *Appl Anim Behav Sci*. 1986;15: 351–364.
41. Newberry RC. [Exploratory behaviour of young domestic fowl](#). *Applied Animal Behaviour Science*. 1999;63: 311–321.
42. Panksepp J. [Affective Neuroscience: The Foundations of Human and Animal Emotions \(Series in Affective Science\)](#). 1 edition. Oxford University Press; 2004.
43. Landgraf D, Long J, Der-Avakian A, Streets M, Welsh DK. [Dissociation of learned helplessness and fear conditioning in mice: a mouse model of depression](#). *PLoS One*. 2015;10: e0125892.
44. Maier SF. [Learned helplessness and animal models of depression](#). *Prog Neuropsychopharmacol Biol Psychiatry*. 1984;8: 435–446.
45. Olsson IAS, J Nicol C, Niemi SM, Sandøe P. [From Unpleasant to Unbearable- Why and How to Implement an Upper Limit to Pain And Other Forms of Suffering in Research with Animals](#). *ILAR J*. 2020. doi:10.1093/ilar/ilz018
46. Job RFS. [Learned helplessness in chickens](#). *Anim Learn Behav*. 1987;15: 347–350.
47. Baxter MR. [The welfare problems of laying hens in battery cages](#). *Vet Rec*. 1994;134: 614–619.
48. Steptoe A, Poole L. [Control and stress](#). In: Fink G, editor. *Handbook of stress: Vol 1 Stress: Concepts, cognition, emotion, and behavior*. Elsevier Academic Press; 2016. pp. 73–80.
49. Jones BR. [Fear and adaptability in poultry: insights, implications and imperatives](#). *World's Poultry Science Journal*. 1996;52: 131–174.
50. Jones RB, Beuving G, Blokhuis HJ. [Tonic immobility and heterophil/lymphocyte responses of the domestic fowl to corticosterone infusion](#). *Physiol Behav*. 1988;42: 249–253.
51. Beuving G, Jones RB, Blokhuis HJ. [Adrenocortical and heterophil/lymphocyte responses to challenge in hens showing short or long tonic immobility reactions](#). *Br Poult Sci*. 1989;30: 175–184.
52. Pértille F, Brantsæter M, Nordgreen J, Coutinho LL, Janczak AM, Jensen P, et al. [DNA methylation profiles in red blood cells of adult hens correlate with their rearing conditions](#). *J Exp Biol*. 2017;220: 3579–3587.
53. Hrabcakova P, Bedanova I, Voslarova E, Pistekova V, Vecerek V. [Evaluation of tonic immobility in common pheasant hens kept in different housing systems during laying period](#). *Arch Anim Breed*. 2012;55: 626–632.

54. A. Fouad M, H. Abdel R A, M. Badawy ES. [Broilers Welfare and Economics under Two Management Alternatives on Commercial Scale](#). *Int J Poult Sci*. 2008;7: 1167–1173.
55. Panksepp J. [Chapter 8 Fear and anxiety mechanisms of the brain: Clinical implications](#). In: Bittar EE, Bittar N, editors. *Principles of Medical Biology*. Elsevier; 2000. pp. 155–177.
56. Duncan IJH. [Science-based assessment of animal welfare: farm animals](#). *Rev Sci Tech*. 2005;24: 483–492.
57. Nicol CJ, Bouwsema J, Caplen G, Davies AC, Hockenhull J, Lambton SL, et al. [Farmed Bird Welfare Science Review](#). Department of Economic Development, Jobs, Transport and Resources, Victoria, Australia; 2017.
58. Bryan Jones R, Faure JM. [Tonic immobility \(“righting time”\) in laying hens housed in cages and pens](#). *Applied Animal Ethology*. 1981;7: 369–372.
59. Shimmura T, Hirahara S, Azuma T, Suzuki T, Eguchi Y, Uetake K, et al. [Multi-factorial investigation of various housing systems for laying hens](#). *Br Poult Sci*. 2010;51: 31–42.
60. Hansen I, Braastad BO, Storbrå J, Tofastrud M. [Differences in Fearfulness Indicated by Tonic Immobility Between Laying Hens in Aviaries and in Cages](#). *Anim Welf*. 1993;2: 105–112.
61. Campbell DLM, Dickson EJ, Lee C. [Application of open field, tonic immobility, and attention bias tests to hens with different ranging patterns](#). *PeerJ*. 2019;7: e8122.
62. Colson S, Michel V, Arnould C. [Welfare of laying hens housed in cages and in aviaries: what about fearfulness?](#) *ArchGeflügelk*. 2006;261: 269.
63. Brantsæter M, Nordgreen J, Rodenburg TB, Tahamtani FM, Popova A, Janczak AM. [Exposure to Increased Environmental Complexity during Rearing Reduces Fearfulness and Increases Use of Three-Dimensional Space in Laying Hens \(Gallus gallus domesticus\)](#). *Front Vet Sci*. 2016;3: 14.
64. Brantsæter M, Tahamtani FM, Moe RO, Hansen TB, Orritt R, Nicol C, et al. [Rearing Laying Hens in Aviaries Reduces Fearfulness following Transfer to Furnished Cages](#). *Front Vet Sci*. 2016;3: 13.
65. Campbell DLM, de Haas EN, Lee C. [A review of environmental enrichment for laying hens during rearing in relation to their behavioral and physiological development](#). *Poult Sci*. 2019;98: 9–28.
66. Campderrich I, Nazar FN, Wichman A, Marin RH, Estevez I, Keeling LJ. [Environmental complexity: A buffer against stress in the domestic chick](#). *PLoS One*. 2019;14: e0210270.

A blueprint for the comparative analysis of pain

67. Fraser D, Duncan IJH. "Pleasures", 'Pains' and Animal Welfare: Toward a Natural History of Affect. *Anim Welf*. 1998;7: 383–396.
68. Murray CJ. Quantifying the burden of disease: the technical basis for disability-adjusted life years. *Bull World Health Organ*. 1994;72: 429–445.
69. Gloor L. *The Case for Suffering-Focused Ethics*. Center on Long-Term Risk; 2016 Aug.
70. Ross M. *Hens with Benefits: Enrichments' Effects on Resilience in Laying Hens*. Mason G, editor. PhD, University of Guelph. 2018.
71. Green TC, Mellor DJ. Extending ideas about animal welfare assessment to include "quality of life" and related concepts. *N Z Vet J*. 2011;59: 263–271.
72. Mellor DJ. Updating Animal Welfare Thinking: Moving beyond the "Five Freedoms" towards "A Life Worth Living." *Animals (Basel)*. 2016;6.
73. Farm Animal Welfare Council. *Farm Animal Welfare in Great Britain: Past, Present and Future*. FAW; 2009.
74. Yeates J. Better to have lived and lost - the concept of a life worth living. In: Butterworth A, editor. *Animal Welfare in a Changing World*. CAB International; 2018. pp. 162–170.
75. Gentle MJ, Corr SA. Endogenous analgesia in the chicken. *Neurosci Lett*. 1995;201: 211–214.
76. Gentle MJ, Tilston VL. Reduction in peripheral inflammation by changes in attention. *Physiol Behav*. 1999;66: 289–292.
77. Wylie LM, Gentle MJ. Feeding-induced tonic pain suppression in the chicken: reversal by naloxone. *Physiol Behav*. 1998;64: 27–30.
78. Gentle MJ. Sodium urate arthritis: effects on the sensory properties of articular afferents in the chicken. *Pain*. 1997;70: 245–251.
79. Gentle MJ, Hunter LN. Neurogenic inflammation in the chicken (*Gallus gallus var domesticus*). *Comp Biochem Physiol C*. 1993;105: 459–462.
80. Eccleston C, Crombez G. Pain demands attention: a cognitive-affective model of the interruptive function of pain. *Psychol Bull*. 1999;125: 356–366.
81. Crombez G, Viane I, Eccleston C, Devulder J, Goubert L. Attention to pain and fear of pain in patients with chronic pain. *J Behav Med*. 2013;36: 371–378.
82. Lumley MA, Cohen JL, Borszcz GS, Cano A, Radcliffe AM, Porter LS, et al. Pain and emotion: a biopsychosocial review of recent research. *J Clin Psychol*. 2011;67: 942–968.

83. Bukola IM, Paula D. [The Effectiveness of Distraction as Procedural Pain Management Technique in Pediatric Oncology Patients: A Meta-analysis and Systematic Review](#). *J Pain Symptom Manage*. 2017;54: 589–600.e1.
84. Ibitoye BM, Oyewale TM, Olubiye KS, Onasoga OA. [The use of distraction as a pain management technique among nurses in a North-central city in Nigeria](#). *International Journal of Africa Nursing Sciences*. 2019;11: 100158.
85. Gentle MJ. [Attentional Shifts Alter Pain Perception in the Chicken](#). *Anim Welf*. 2001;10: 187–194.
86. USDA. [Layers 2013 Part I: Reference of Health and Management Practices on Table-Egg Farms in the United States](#). USDA; 2014.
87. Bell DD. [Historical and current molting practices in the U.S. table egg industry](#). *Poult Sci*. 2003;82: 965–970.
88. Anderson KE. [Molting of laying hens: test results from North Carolina and implications for US and German egg producers](#). *LOHMANN Information*. 2016;50: 12–17.
89. Rufener C, Baur S, Stratmann A, Toscano MJ. [A Reliable Method to Assess Keel Bone Fractures in Laying Hens From Radiographs Using a Tagged Visual Analogue Scale](#). *Front Vet Sci*. 2018;5: 124.
90. Stratmann A, Fröhlich EKF. [Genetic selection to increase bone strength affects prevalence of keel bone damage and egg parameters in commercially housed laying hens](#). *Poult Sci*. 2016.
91. Fernyhough M, Nicol CJ, van de Braak T, Toscano MJ, Tønnessen M. [The Ethics of Laying Hen Genetics](#). *J Agric Environ Ethics*. 2019.
92. Greger M. [Bird Flu: A Virus of Our Own Hatching](#). Edición: 1. Lantern Books,US; 6 de diciembre de 2006.
93. Schuck-Paim C, Alonso WJ. [Pandemics, global health and consumer choices](#). Cria editora. 2020.
94. Greger M. [The human/animal interface: emergence and resurgence of zoonotic infectious diseases](#). *Crit Rev Microbiol*. 2007;33: 243–299.
95. Nicol C. [Feather pecking and cannibalism](#). In: Mench JA, editor. *Advances in Poultry Welfare*. 2018. pp. 175–197.
96. FAWC. [Opinion on osteoporosis and bone fractures in laying hens](#). Farm Animal Welfare Council; 2010.